

## **PYTHON PROJECT**

**Title: Advanced Quiz Game with Score and Hints using Recursion**

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### **AIM**

To design and develop an advanced Quiz Game using Python without loops, utilizing recursion, scoring system, and hints for better user interaction.

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### **OBJECTIVE**

- To implement recursion instead of loops
  - To design a quiz-based interactive application
  - To maintain score dynamically
  - To use lists and dictionaries for storing questions
  - To apply conditional statements for answer checking
  - To enhance user experience with hints and feedback
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### **PROCEDURE**

1. Open Python IDLE or any editor and create a file named quiz\_game.py.
2. Create a list of dictionaries where each dictionary contains:
  - Question
  - Correct answer
  - Hint
3. Initialize variables:

- score to track user performance
  - index to track current question
4. Define a recursive function `play_quiz()` that takes:
    - List of questions
    - Current index
    - Current score
  5. Inside the function:
    - Check if all questions are completed
    - If yes, display final score and stop recursion
  6. Display the current question to the user.
  7. Ask the user to enter their answer.
  8. Use conditional statements to:
    - Check if the answer is correct
    - Increase score if correct
    - Show hint if wrong
  9. If wrong:
    - Give one more chance using recursion
  10. Move to the next question by calling the function again with updated index.
  11. Ensure recursion continues until all questions are completed.
  12. Display final score and performance message.
  13. Call the function from main program.
  14. Test the program with multiple inputs.
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## **PYTHON CODE**

```
def play_quiz(questions, index, score):
```

```
    # Base condition: end of quiz
```

```
if index == len(questions):  
    print("\n🎉 Quiz Completed!")  
    print("Final Score:", score, "/", len(questions))
```

```
if score == len(questions):  
    print("Excellent! 🏆")  
elif score >= len(questions)//2:  
    print("Good Job 🌟")  
else:  
    print("Keep Practicing 📖")  
return
```

```
q = questions[index]
```

```
print("\nQuestion", index + 1)  
print(q["question"])
```

```
answer = input("Your Answer: ").lower()
```

```
if answer == q["answer"]:  
    print(" 🟢 Correct!")  
    play_quiz(questions, index + 1, score + 1)
```

```
else:  
    print("🔴 Wrong!")  
    print("Hint:", q["hint"])
```

```
retry = input("Try again? (yes/no): ").lower()
```

```
if retry == "yes":  
    retry_answer = input("Your Answer: ").lower()
```

```
if retry_answer == q["answer"]:  
    print(" 🟢 Correct!")  
    play_quiz(questions, index + 1, score + 1)  
else:  
    print("🔴 Still Wrong! Moving to next.")
```

```
        play_quiz(questions, index + 1, score)
    else:
        play_quiz(questions, index + 1, score)
```

```
def main():
```

```
    print("🎮 Welcome to Advanced Quiz Game\n")
```

```
    questions = [
```

```
        {
            "question": "What is the capital of India?",
            "answer": "delhi",
            "hint": "It starts with D"
        },
```

```
        {
            "question": "Which language is used for AI?",
            "answer": "python",
            "hint": "Popular programming language"
        },
```

```
        {
            "question": "2 + 2 * 2 = ?",
            "answer": "6",
            "hint": "Follow BODMAS rule"
        },
```

```
        {
            "question": "Who is known as the father of computers?",
            "answer": "charles babbage",
            "hint": "His first name is Charles"
        }
    ]
```

```
    play_quiz(questions, 0, 0)
```

```
main()
```

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## OUTPUT

```
🎮 Welcome to Advanced Quiz Game

Question 1
What is the capital of India?
Your Answer: delhi
✅ Correct!

Question 2
Which language is used for AI?
Your Answer: java
❌ Wrong!
Hint: Popular programming language
Try again? (yes/no): yes
Your Answer: python
✅ Correct!

Question 3
2 + 2 * 2 = ?
Your Answer: 8
❌ Wrong!
Hint: Follow BODMAS rule
Try again? (yes/no): 6

Question 4
Who is known as the father of computers?
Your Answer: charles babbage
✅ Correct!

🎉 Quiz Completed!
Final Score: 3 / 4
Good Job 🍀

=== Code Execution Successful ===
```

## RESULT

The Advanced Quiz Game was successfully implemented using recursion without loops. The program handled multiple questions, scoring, hints, and user interaction effectively.